RageSense: Leveraging Acoustic Sensing and LLM-Based Intervention for Emotion Regulation in Mobile Gaming

Cong Liu*
South China University of Technology
Guangzhou, China
202420163862@mail.scut.edu.cn

Ruihao Zheng*
East China Normal University
Shanghai, China
894945409@qq.com

Junbin Ren
East China Normal University
Shanghai, China
51265901135@stu.ecnu.edu.cn

Kaiyi Guo
School of Software
Shanghai Jiao Tong University
Shanghai, China
gky2023@sjtu.edu.cn

Qian Zhang
RFID and IOT lab
Shanghai Jiao Tong University
Shanghai, China
qwert3472@sjtu.edu.cn

Dong She
South China University of Technology
Guangzhou, China
ftdshe@mail.scut.edu.cn

Yuting Bai
Design
South China University of Technology
GuangZhou, GuangDong, China
202321055492@mail.scut.edu.cn

Zhanpeng Jin
School of Future Technology
South China University of Technology
Guangzhou, Guangdong, China
zjin@scut.edu.cn

Yang Gao†
School of Future Technology
South China University of Technology
Guangzhou, China
gaoyang@cs.ecnu.edu.cn

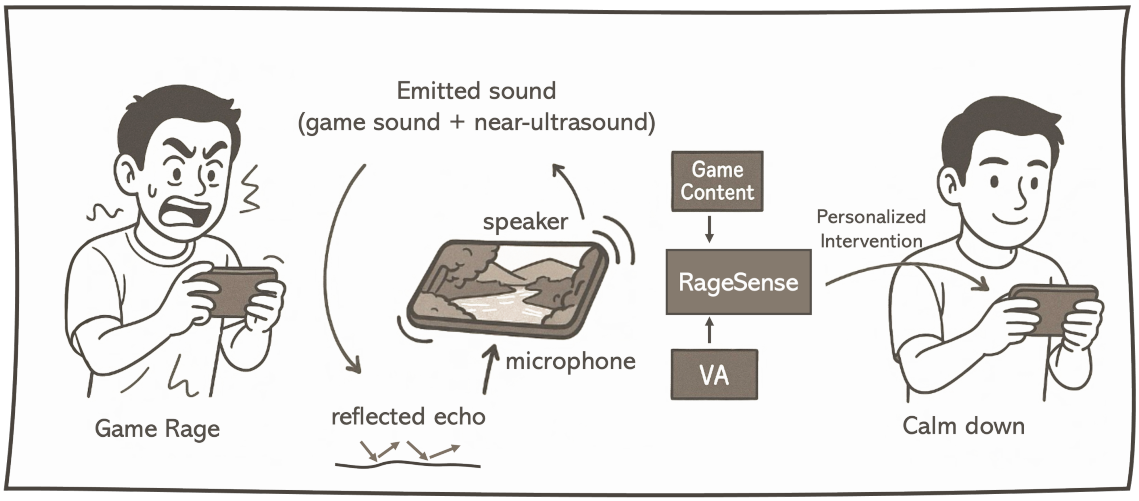

Figure 1: By emitting inaudible soundwaves alongside game audio, RageSense captures facial reactions through reflected echoes, providing a non-invasive way to estimate player emotions. This data, combined with game context, drives personalized interventions powered by a large language model (LLM), effectively reducing player rage and enhancing gameplay experience.

Abstract

RageSense introduces a novel system for detecting and regulating player frustration during mobile gaming. Instead of relying on coarse emotion labels, RageSense estimates users' valence and arousal levels in real time using near-ultrasonic acoustic sensing. By analyzing facial muscle movements via built-in smartphone speakers and microphones, our approach enables emotion sensing without requiring cameras or wearables, constituting a more unobtrusive, environment-resilient, and privacy-friendly approach than traditional emotion recognition. To transform detection into action, we integrate a large language model (LLM) that generates empathetic, context-aware interventions based on gameplay screenshots, behavioral signals, and emotional trajectories. These interventions are delivered in real time, tailored to the user's emotional state, and designed to mitigate rage while enhancing player well-being. In a 53-participant field study, our system improved emotional

*Both authors contributed equally to this research.
†Yang Gao is the corresponding author

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state immediately after triggers and was preferred over random or template-based messages. To our knowledge, this is the first demonstration of near-ultrasonic, on-phone valence-arousal regression during mobile gameplay that directly drives real-time, context-aware interventions.

CCS Concepts

• **Human-centered computing → Interaction techniques**.

Keywords

Affective Computing, Acoustic Sensing, Mobile Gaming, Emotion Regulation

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1 Introduction

Mobile gaming has become a dominant form of digital entertainment, accounting for nearly 57% of the global video game revenue[1]. Players engage in mobile games for various reasons, including relaxation, social interaction, and competition. However, along with the rapid growth of mobile gaming, concerns have emerged regarding its impact on players' emotions and behaviors. Many players experience negative emotions such as frustration, disappointment, and anger due to in-game failures, network issues, or competitive pressure. If left unregulated, these emotions may lead to problematic behaviors such as excessive gaming, cyberbullying, and even real-world emotional distress. Existing research has demonstrated that players who enter a "rage loop" often struggle to disengage from the game [4]. For instance, after consecutive failures, some players refuse to quit and instead prolong their play sessions in an attempt to "take revenge," thereby exacerbating negative emotional cycles. This compulsive engagement not only affects mental well-being but also contributes to gaming addiction.

Current interventions for in-game toxicity primarily rely on reactive measures such as chat filters, along with block, mute, and report functions.[69, 78] While these tools can shield the community from overt hostility, they have critical limitations. First, they are reactive, addressing toxic behavior only after an angry outburst has already occurred, rather than preventing the underlying emotional escalation. Second, these methods are designed to protect others, but they do not help calm the frustrated player; in fact, being blocked or reported may even aggravate their negative feelings. Other approaches, like generic, time-based reminders to "take a break,"[57]are often emotion-agnostic, failing to engage with the player's actual mental state and the specific in-game events that trigger their frustration.

Our work introduces a proactive and non-disruptive intervention framework tailored for fast-paced games where pausing is not a viable option. We address the core challenge of designing an intervention that is perceived as supportive rather than intrusive. To achieve this, we propose a system that leverages a Large Language Model (LLM) to provide timely, personalized support upon detecting the onset of player frustration. Unlike previous systems, our LLM utilizes real-time game events to generate concise, customized messages delivered via an unobtrusive on-screen overlay. This context-aware approach ensures the feedback is directly relevant to the player's situation, offering empathetic guidance over generic advice to help players regulate their emotions more effectively.

Real-time emotion detection is essential for proactive interventions. Traditional methods, such as camera-based facial analysis [22, 27] and speech analysis [34, 63], face challenges in mobile environments due to lighting variability, occlusion, and face significant privacy risks. Recent studies have explored ultrasonic sensing to address these issues. Wearable acoustic systems such as EarIO [38] and EyeEcho [37] demonstrate that ultrasonic signals can track fine-grained facial movements in real time. However, these systems require earable or glasses-mounted hardware and assume stable head–sensor geometry, making them unsuitable for mobile gaming, where users frequently change posture and do not wear additional devices. Moreover, these systems focus on expression classification or geometric motion tracking, rather than modeling continuous valence–arousal dynamics. Recent mobile-based acoustic systems like ExpresSense [24], UFace [68], and EARFace [79] recognize discrete facial expressions but are less suited for tracking rapid, subtle emotional transitions that characterize gameplay. They also do not address practical challenges unique to mobile gaming—such as two-handed screen interactions, which generate severe acoustic artifacts and corrupt the facial echo signal.

We present RageSense, which bridges these gaps by enabling continuous valence–arousal estimation directly from a standard smartphone, requiring no additional hardware, and by introducing an IMU-informed filtering mechanism to suppress acoustic corruption caused by intense touch interactions. This system integrates near-ultrasonic emotion detection with large language model (LLM)-generated interventions to support emotion regulation in mobile gaming. The key contributions of this work include:

- Continuous Valence-Arousal Estimation for Game Environments: We introduce a novel approach to emotion recognition in mobile gaming by being the first (to our best knowledge) to map acoustic spectrograms to continuous valence-arousal space. This approach is enhanced by custom ultrasonic signals that reduce interference from smartphone artifacts, as well as the integration of IMU data to detect excessive finger movements, ensuring better spectrogram quality and more accurate valence-arousal estimation.
- Personalized, Context-Aware Interventions Based on Large Language Models: Our system integrates emotion detection with large language model (LLM)-powered interventions, providing context-sensitive, personalized feedback based on real-time game content and the player's emotional state. This tailored approach fosters more empathetic and effective emotional regulation during gameplay.
- Comprehensive In-Field Study Evaluation: We conducted an in-field study with 53 participants over three weeks to evaluate the effectiveness of our emotion regulation system. Compared to baseline methods, our system demonstrated

significant improvements in emotional regulation, providing clear evidence of its effectiveness in real-world gaming scenarios.

The remainder of this paper is structured as follows. We first review related work and present a formative study on player needs (Sec. 2-3). We then detail the design and implementation of RageSense, our system for emotion recognition and intervention, covering its sensing, prediction, and intervention modules (Sec. 4-6). Subsequently, we report on an in-field study evaluating the effectiveness of its real-time interventions (Sec. 7). Finally, we discuss the implications and limitations of our work and conclude the paper (Sec. 8-9).

2 Background and Related Work

We first summarize related work about facial emotion recognition methods, as well as smartphone-based emotion recognition techniques. Then, we summarize existing research about user engagement and emotions in the context of video games.

2.1 Negative Emotion in Games

Emotion recognition in gaming has become an increasingly important area of research as it holds the potential to create more adaptive, personalized gaming experiences[12, 28, 39, 62, 71]. The goal is to detect players' emotional states in real-time, allowing games to respond dynamically and maintain engagement by adjusting difficulty, providing feedback, or offering interventions[31, 33, 67, 83]. Traditional emotion recognition methods have predominantly relied on either physiological signals or behavioral cues, such as facial expressions and voice analysis. These methods, however, have their own set of challenges, particularly when applied to interactive gaming contexts[10, 15, 66].

Early research focused primarily on physiological sensing, such as heart rate, skin conductance, and EEG, to monitor emotions. These physiological signals provide valuable insights into arousal levels and emotional states, as they are directly tied to the autonomic nervous system. Studies have demonstrated the ability to predict player emotions, such as frustration, excitement, or stress, using these signals. For instance, Yannakakis et al[39]. developed a system that integrated physiological data from sensors to detect discrete emotional states like frustration and enjoyment during gameplay. Similarly, Mishra et al[43]. used facial emotion recognition techniques to detect the emotional states of players during video game interactions.

2.2 Emotion Regulation and In-Game Interventions

Given the increasing emotional nature of video games, recently, more and more research efforts have shown that video games would affect the emotional experience of players [7, 19]. Video games offer players diverse emotional experiences, ranging from positive emotions, such as happiness and satisfaction, to intense negative emotions, like sadness and anger.

Many studies also proposed to understand the types of sensations elicited by these game products[13, 14, 17, 40, 50, 53]. These explorations are based mainly on two approaches: model-based and model-free. The first group uses a top-down approach, building the model on a theoretical framework that maps the players' affect. For example, a model can be based on the game emotion theory [32, 64], or a general emotion theory [48], such as the definition of a set of parameters that describes the players' behavior. In [54], the researchers studied the players' satisfaction with the game rewards (Looting System) in Massive Multiplayer Online (MMO) and Multiplayer Online Battle Arena (MOBA) games through a set of simulations. The second approach (bottom-up) infers the players' emotions mapping them without the use of any particular theoretical framework. This type of players' analysis can be performed in two ways: studying the players' behaviors in video games, or using physiological information to infer players' emotions.

Many techniques are developed to analyze and detect the emotions of players. Mishra *et al.* [43] proposed a Hidden Markov Model (HMM) based real-time player emotional recognition system using recorded videos to capture player's facial emotions. Granato *et al.* [18] proposed a hardware and software setup to collect users' physiological data (e.g., EMG, EDA) and then asses players' emotions during video games. Another study [41] explored emotion recognition and inhibitory control for adolescent players during violent games.

2.3 Ubiquitious Sensing for Emotion Detection

It is widely accepted that emotional changes are accompanied by both behavioral reactions (e.g., facial expressions, speech) and physiological responses (e.g., heart rates) [21, 23, 29, 52, 60, 65, 72, 77, 81]. However, directly measuring physiological signals with dedicated sensors such as EEG is often impractical for everyday contexts like gaming [6, 16, 25, 49]. Consequently, with smartphones becoming central to social interaction and entertainment, researchers have turned to their built-in sensors, observing that smartphone usage is closely associated with user affective experiences. For instance, prior work has inferred user's mood from social media interaction patterns [76] or predicted emotions using data from motion, touch, and eye-tracking sensors during social network use [55]. Among these, motion sensors are particularly effective due to their ubiquity, low cost, and the richness of the movement data they capture, making them a widely used vehicle for detecting emotions.

Facial expressions are a primary indicator of emotion, but traditional camera-based methods face challenges like lighting changes and occlusion. To overcome these, acoustic sensing offers a promising alternative. Systems such as EarIO [38] and EyeEcho [37] introduce wearable acoustic sensing that can continuously track facial expressions. EarIO, deployed in earables, and EyeEcho, mounted on glasses, emit acoustic signals towards the face to capture subtle skin deformations, providing continuous real-time tracking. Kar *et al.* [24] introduced ExpresSense, a system that uses near-ultrasound signals from smartphones to detect facial expressions without a camera. Similarly, Wang *et al.* [68] developed UFace, an acoustic system recognizing seven typical facial expressions. These systems show the feasibility of acoustic sensing for facial expression recognition, focusing on discrete emotion classification.

In contrast, our work focuses on modeling emotion in the continuous dimensions of valence and arousal, better capturing the dynamic nature of emotions in gaming. Discrete labels oversimplify evolving emotional states, while continuous modeling reflects how

emotions change over time. The valence-arousal model offers a fine-grained representation of emotional experiences, enabling richer descriptions beyond static categories. Russell [56] introduced this framework, forming the foundation for affective modeling. Previous work [11, 45, 47, 70, 82] shows the value of continuous emotion prediction across modalities. Our work extends this by applying the valence-arousal model to mobile gaming and integrating acoustic sensing for real-time emotion tracking and adaptive interventions.

3 Formative Study

To deeply understand players' negative emotional experiences during mobile gameplay and to identify challenges in implementing an effective intervention system, we conducted focus group discussions and semi-structured interviews with typical mobile game players. The goal of this qualitative study was to: 1) identify common triggers and manifestations of negative emotions, especially frustration and rage, during gameplay; 2) explore players' perceptions of their own emotional reactions and existing coping mechanisms; and 3) understand practical challenges players perceive in adopting technology-driven emotion-regulation interventions.

3.1 Participants and Methodology

We recruited 24 active mobile gamers (M=24.2, SD=2) from a local university community. Participants played for at least one hour daily for entertainment. Crucially, eligibility was determined through a screening survey in which participants indicated the frequency of intense negative emotions experienced during gaming over the past month and provided valence–arousal (1–5) ratings and brief descriptions of their most recent episode. Only individuals who reported experiencing intense frustration or rage at least once per week were enrolled, ensuring alignment with our target demographic.

Our methodology employed a two-stage qualitative approach. We first conducted two 60-minute focus groups to uncover shared experiences and common themes surrounding frustrating gameplay. These were followed by individual semi-structured interviews to probe deeper into personal emotional responses and to gather specific perceptions regarding the acceptance, usability, and privacy concerns of potential technology-based interventions. All sessions were moderated by two experienced researchers and audio-recorded for subsequent thematic analysis.

3.2 Understanding Negative Emotions Intervention in Mobile Gaming

DQ1: How to design an emotion recognition system that players would actually trust and adopt?

Our formative study revealed a key tension: while the vast majority of players (95.8%) were open to receiving in-game emotional support, their acceptance hinged on addressing three key concerns: data privacy(20/24), with users opposing the storage of personally identifiable data like facial or audio recordings ; emotional trajectory(18/24): Participants questioned whether the system could capture the dynamic nature of their in-game emotions in real-time; and performance overhead(15/24), fearing that sensing would consume critical system resources and introduce lag .

Elaborating on these concerns, players were skeptical of systems that reduce complex emotions to static labels. They argued that the timing of an intervention is critical. For instance, one participant vividly expressed this concern: *"What if the system tries to 'help' me based on my anger from the last lost team fight, right when we've just won a new one? That would be incredibly annoying."* Similarly, the concern about performance overhead was a deal-breaker. Another participant stated unequivocally, *"If this system makes my phone hot or the game lag, I'll turn it off immediately. Smooth gameplay comes first."* These insights established two core design requirements: the system must not only track the temporal dynamics of emotion but also operate with negligible performance impact. This directly motivated our choice to combine lightweight acoustic sensing with a continuous valence-arousal model, as this approach is well-suited for tracking real-time emotional shifts without compromising game performance.

DQ2: How to deliver an intervention that feels genuinely helpful, not just another distraction?

Our second design principle guides the intervention's content and delivery. A significant 87.5% of participants stressed the need for personalization, a need driven by specific concerns about how a generic system could fail. Their concerns centered on three potential failures: improper timing (17/24), contextual mismatch (14/24), and a patronizing tone (10/24).

These concerns were vividly illustrated by participants. On improper timing, one player said, *"If I'm concentrating on a gacha pull and the system distracts me, I would get really angry."* The issue of contextual mismatch was equally critical, as another explained, *"When we lost a team fight because a teammate didn't show up, the system telling me to 'watch the minimap' would just amplify my frustration."* Finally, a patronizing tone could undermine an expert player's experience: *"I have over a thousand hours on this champion. For the system to tell me 'Don't forget to use your ultimate' after a complex tactical mistake feels like it's treating me like a complete novice."* These examples highlight a critical requirement: an effective system must understand the cause of an emotion, not just its presence. This principle led us to explore solutions capable of deep contextual understanding, making Large Language Models (LLMs) a compelling choice. By interpreting emotional data alongside in-game events, LLMs empower our system to handle these subtleties—delivering feedback that is timely, relevant, and respectful—thereby transforming a potential annoyance into meaningful support.

3.3 Design Considerations for Mobile Game Interventions

Two primary design considerations guide the development of RageSense's emotion detection and intervention. system.

Emotion Estimation. Traditional approaches, like visual-based emotion recognition or behavior analysis through touch interactions, are often hindered by significant limitations. Visual-based methods, while powerful, face privacy concerns and may require cumbersome equipment. Touchscreen behavior models, though intuitive, can be heavily influenced by the game type or the player's hand interactions, making it difficult to consistently map to emotion-related signals like valence and arousal. To address these challenges,

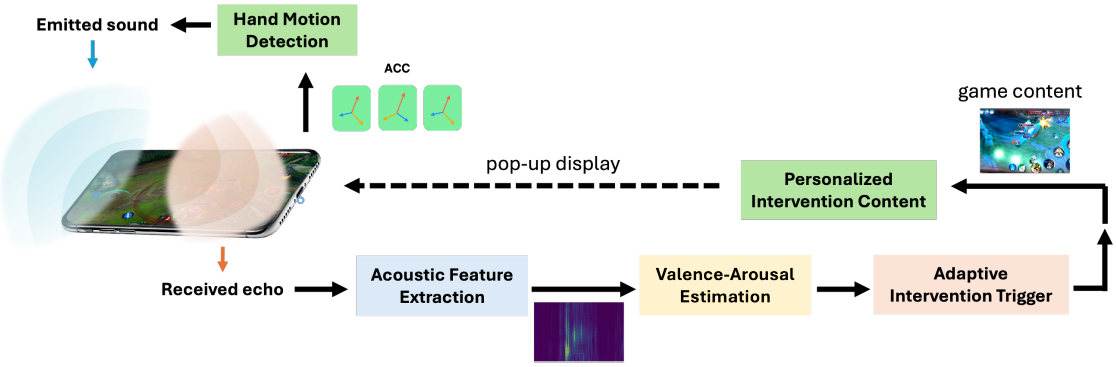


Figure 2: System pipeline of RageSense, showing hand motion detection, acoustic feature extraction, valence-arousal estimation, and adaptive intervention trigger for personalized emotion regulation in mobile gaming.

we adopted an acoustic sensing approach to predict emotions by analyzing reflected echo spectra to estimate valence and arousal. Our approach offers a privacy-friendlier alternative by reducing sensitive data capture: it avoids recording facial images and background scenes; instead, it relies on non-semantic micro-reflection signals tied to subtle facial dynamics and hand movements. This makes it a discreet and minimally intrusive modality. Additionally, it is less affected by external factors like game type or player behavior, ensuring consistent emotion detection.

Intervention delivery. As discussed in Design Consideration 2, personalized interventions are essential to making emotional support in games more effective. To address this, the intervention design in RageSense leverages both emotion detection and the understanding capabilities of large language models (LLMs) to deliver tailored, context-sensitive feedback. By analyzing the player's emotional state in real time alongside game-specific context (such as a screenshot of the current game state), RageSense generates personalized intervention content.

As shown in Figure 2, we implemented RageSense, which consists of two integrated modules: the Acoustic-based Emotion Estimation Module, and the LLM-based Personalized Intervention Module. These modules enable precise and context-aware emotion detection and effective real-time interventions, thereby providing a robust and user-friendly experience. Figure 2 illustrates the architecture and the workflow of the RageSense system. We detail each module in the following sections.

4 RageSense Design

4.1 Acoustic Sensing-based Emotion Estimation

In this section, we propose an emotion recognition system using acoustic sensing with the built-in speakers and microphones of smartphones. The main technical challenges are recognizing facial expressions in gaming environments, minimizing the interference of acoustic sensing on the gameplay, and mapping acoustic features to continuous valence and arousal values. To address these challenges, we have designed three key modules: finger motion removal, an acoustic sensing pipeline, and a neural network for valence-arousal value estimation, shown in Figure 4.

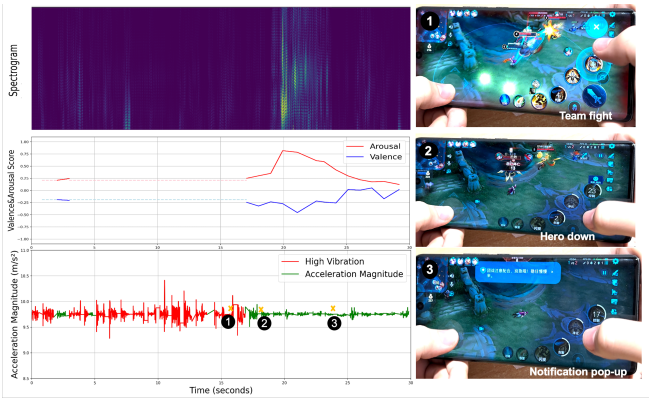


Figure 3: Emotion-aware acoustic sensing during mobile gaming. (1) spectrogram from acoustic signals; (2) valence-arousal dynamics; (3) IMU signal variations; and (4) gameplay scenes for combat, failure, and emotion-driven intervention. Acoustic sensing is enabled only when IMU data indicates device stability, ensuring reliable features for on-device emotion recognition.

4.1.1 IMU-Gated Motion Removal. In highly engaging or emotionally charged gaming scenarios, players often exhibit intense hand and finger movements, such as rapid screen tapping or swiping, which can introduce significant motion-induced noise in the acoustic sensing signals. These movements can interfere with the accuracy of emotion detection based solely on acoustic features. To address this challenge, we integrate a motion-aware approach that leverages data from the Inertial Measurement Unit (IMU) sensors, including accelerometers and gyroscopes, to identify the echo frames corrupted by aggressive taps/swipes.

Large-scale hand gestures, such as aggressive swiping or tapping, produce distinct patterns in the IMU data, which can be used to differentiate between minor and significant hand operations. To ensure accurate emotion detection, our system distinguishes between two scenarios based on IMU data:

- Significant Hand Operations: where the IMU amplitude exceeds a predefined threshold, the system temporarily suspends acoustic sensing to avoid distortion from motion noise.

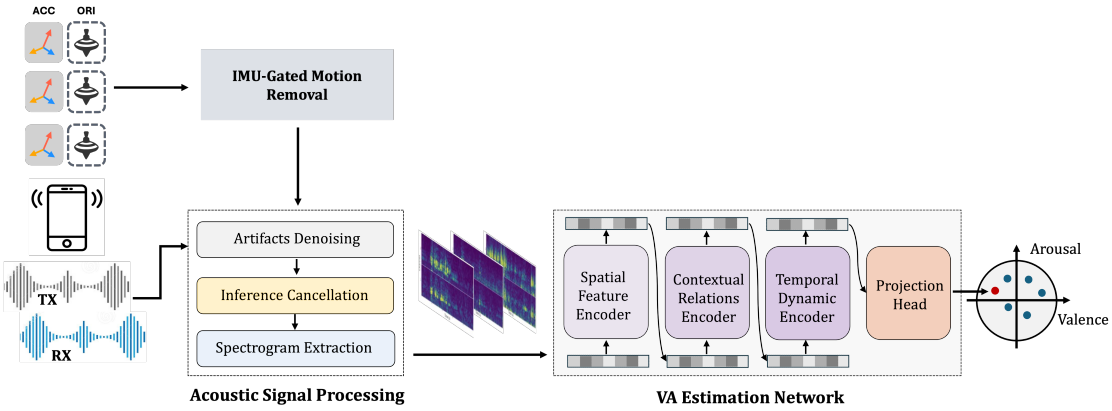


Figure 4: VA-based emotion estimation pipeline.

- Minor Hand Operations: when the IMU amplitude remains low, indicating minimal interference with the acoustic signals, the system continues to rely on the acoustic features for emotion detection.

To detect and isolate significant hand movements, we employ an energy-based approach to analyze the vibrations generated by hand operations. Specifically, we combine accelerometer and gyroscope magnitudes to produce a unified vibration magnitude. Next, the short-time energy is calculated by accumulating the squared values of the combined signal within a sliding window. A threshold, set to the mean plus 0.25 times the standard deviation of the energy, is used to identify the onset of high-energy segments. As illustrated in Figure 3, this approach effectively isolates segments of intense hand movement during mobile gameplay, thereby preventing their interference with the subsequent acoustic feature extraction

For passive sensing applications, such as finger tapping, we set the window length to 0.1 seconds, which covers the duration of typical passive vibration signals. After segmenting the vibration signals, they are normalized relative to their maximum amplitude to account for variations in the intensity of taps. This ensures that the system can effectively differentiate between noise generated by minor operations and genuine signals associated with significant emotional responses.

4.1.2 *Acoustic Signal Processing.*

Frequency Modulation Artifacts Denoising. In near-ultrasonic Frequency Modulated Continuous Wave (FMCW) systems, particularly in mobile gaming contexts, the modulation process can introduce low-frequency variations or gaps due to non-linearities in the frequency sweep. These can result in audible low-frequency noise, impacting the user experience, especially during gaming sessions. To address these, we applied a frequency modulation artifact denoising technique that smooths the FMCW signal. This involves using a raised cosine window to smooth the beginning and end of the signal, reducing abrupt changes. The method works by multiplying the original signal with the raised cosine function at both ends, ensuring smoother transitions and minimizing audible noise.

Interference Cancellation. In the dynamic setting of mobile gaming, we leverage the smartphone's own speaker to emit ultrasonic signals, transforming it into an active sensor for capturing facial expressions. The received signal, however, contains both the desired expression-related features and strong reflections from static objects, such as tables and chairs. To suppress these static reflections, we compute the difference between consecutive spectrograms, which cancels out the stationary background and isolates subtle facial signals. In addition, since both the speaker and microphone are integrated within the same device, part of the emitted signal can directly couple into the microphone, introducing interference with a fixed delay. Inspired by the interference cancellation technique in works [36–38], we estimate this constant delay using Time Alignment and subtract the known artifact during signal processing, effectively eliminating the direct-path interference and focusing on reflective signals from the user's face.

4.1.3 *Feature Extraction.*

Our feature extraction pipeline, illustrated in Figure 5, transforms subtle physical deformations of the user's face into a digital representation. The process begins when the smartphone speaker emits an inaudible ultrasonic chirp signal (TX). As the signal reflects off the user's face, its echo patterns are modulated by facial muscle movements, and these echoes are captured by the phone's dual microphones (RX ch1, RX ch2), creating a continuous two-channel data stream encoding the user's expression.

To decode this information, we compute the cross-correlation between the transmitted (TX) and received (RX) signals. This step reveals the time-varying signature of the facial echoes, converting raw audio into structured correlation sequences. However, body movements or hand gestures can introduce transient noise, reducing signal stability. We apply a Gaussian filter to smooth the correlation data, suppressing fluctuations while preserving facial motion patterns.

Finally, we segment and stack these cleaned correlation sequences frame-by-frame, creating a 2D matrix, termed a spectrogram. Each row represents the acoustic response from a single chirp, forming a spectrogram of the facial expression over time. This spectrogram serves as the final input for our deep learning model.

For our implementation, the transmitted ultrasonic signal has a sampling rate of F_s = 48 kHz. Each chirp spans T_w = 10 ms with a

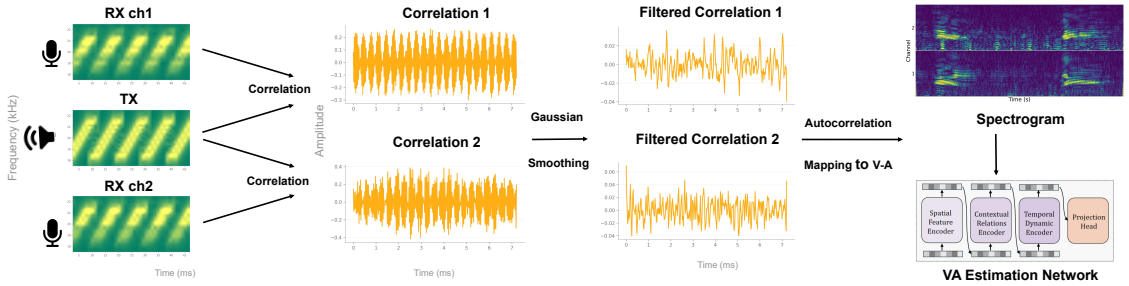


Figure 5: Signal processing pipeline for extracting facial features from ultrasonic reflections.

bandwidth of $B = 4$ kHz (18–22 kHz), ensuring the signal is inaudible and smartphone-compatible. We followed prior acoustic-sensing studies [36–38] and adopted a cross-correlation-based approach to extract features produced by subtle facial motions. The cross-correlation is formally defined as:

$$C(n) = Tx(n) * Rx(n) = \sum_{m=0}^{N-1} Tx(m) \cdot Rx(m+n), \quad n \geq 0 \tag{1}$$

where $Tx(n)$ is the transmitted chirp and $Rx(n)$ is the filtered received signal for each channel.

4.1.4 Valence and Arousal Estimation. To map the acoustic spectrograms into continuous emotional states, we designed the valence-arousal Estimation Network, a lightweight hybrid architecture optimized for real-time mobile deployment. As illustrated in Figure 4, a pipeline of specialized encoders is employed to map the acoustic patterns in the spectrograms into the two-dimensional valence-arousal space.

The process begins with a Spatial Feature Encoder (ResNet-18) that efficiently scans the spectrogram to extract foundational local patterns, which serve as the essential building blocks of an expression. These features are subsequently processed by a Contextual Relations Encoder, a Sparse Transformer, which models long-range dependencies to form a holistic contextual representation with an efficient complexity of $(O(N \log N))$. To capture the temporal dynamics of emotional expression, the contextualized features are then fed into a Temporal Dynamic Encoder (LSTM). The final hidden state from the LSTM is mapped by a Projection Head to predict continuous values for valence and arousal.

To handle the continuous data stream, the system operates on a sliding-window basis. We chose a 5-second window with a 2-second stride following prior valence-arousal-regression studies [75, 84], which showed that a 5-second temporal context is necessary to capture a complete emotional fluctuation. To further enhance the stability of the real-time output, we introduce a standard temporal smoothness loss during training. Following common practice, we penalize rapid frame-to-frame changes in the predicted valence and arousal using a Huber-style difference loss, which is less sensitive to outliers than a simple squared penalty.

For a sequence of length T, the temporal smoothness loss can be written as:

$$l_t = \rho(\hat{V}_t - \hat{V}_{t-1}) + \rho(\hat{A}_t - \hat{A}_{t-1}), \tag{2}$$

where the penalty function $\rho(\cdot)$ is defined as

$$\rho(e) = \begin{cases} \frac{1}{2}\frac{e^2}{\beta}, & \text{if } |e| < \beta, \\ |e| - \frac{1}{2}\beta, & \text{otherwise.} \end{cases} \tag{3}$$

Here, $\hat{V}_t$ and $\hat{A}_t$ denote the predicted valence and arousal at time t, and β ($\beta = 1$) controls the transition between the squared and linear regimes. In practice, we simply aggregate l_t over the sequence using a standard reduction (mean or sum).

Based on extensive experiments, we identified the following configuration as optimal: the ResNet-18 feature extractor; a 4-layer Sparse Transformer encoder with 8 attention heads; and a 2-layer LSTM with a hidden size of 256. This architecture, containing approximately 25M parameters, achieved the best balance between predictive accuracy and on-device efficiency among the variants we tested. We used the AdamW optimizer with a learning rate of 5×10^{-5} and trained the model for 50 epochs before deploying it using the PyTorch Mobile framework.

4.2 LLM-powered Negative Emotion Intervention

4.2.1 Adaptive Intervention Trigger Algorithm. As detailed in Appendix C, the algorithm processes real-time ultrasound-based affective signals to detect emotional shifts and provide context-aware interventions during gameplay. It runs continuously while the session is active. First, an ultrasound spectrogram is acquired, and the system estimates the instantaneous valence $V(t)$ and arousal $A(t)$ using a Transformer+LSTM model. These values are appended to a circular buffer. To smooth fluctuations, an exponential moving average (EMA) is applied to both valence and arousal. Once the buffer is filled, dynamic thresholds are computed to detect significant negative emotional shifts.

The thresholds are based on initial values from "anger/frustration" instances in our dataset ($V_0 = -0.453 \pm 0.181$, $A_0 = 0.480 \pm 0.276$). A 60-second sliding window continuously calculates the player's current mean and standard deviation. The adaptive thresholds for valence T_V and arousal T_A are set as:

$$T_V = \min(V_0 - k\sigma_V, \mu_V - k\sigma_V), \quad T_A = \max(A_0 + k\sigma_A, \mu_A + k\sigma_A) \tag{4}$$

The sensitivity parameter k linearly decays from 1.2 to 0.8 during the session, minimizing false alarms while adapting to recent emotional baselines. A six-player pilot study confirmed this range reduces false alarms and avoids missing true frustration events. An

intervention is triggered when valence falls below T_V and arousal exceeds T_A.

The algorithm uses the CUSUM change-point detection method[35], tracking gradual shifts in valence-arousal. If a change-point is detected, the system checks if the smoothed values exceed dynamic thresholds. If met, screenshots of the current game state are captured for the intervention context.

Next, the system constructs a prompt for the cloud-based LLM, combining valence-arousal history and captured images. The LLM generates a personalized intervention message; if not, a default message is provided. Finally, the intervention is displayed as a semi-transparent pop-up for 5 seconds, ensuring minimal disruption. Our intervention message is an in-app UI component displayed through an Android OS-level overlay window. This mechanism allows RageSense to present messages above the game without accessing or modifying any game code.

4.2.2 LLM-based Content Generation. The intervention module of the RageSense system uses a Large Language Model (LLM) to generate personalized, context-aware messages in response to detected negative emotions. The LLM integrates two main input sources: real-time gameplay screenshots and emotional data (valence-arousal values). Game screenshots capture key aspects of the player's current situation, such as game status, challenges, and performance. This visual data is used to generate targeted recommendations. Meanwhile, valence-arousal data encodes the emotional state, offering direct insight into the player's emotional condition and trends, enhancing the intervention's precision.

The prompt guides the LLM to deliver personalized, context-aware responses by considering the player's emotional state and gameplay scenario. It prioritizes empathy, tailoring responses for comfort in negative emotions and motivation for heightened emotions. The LLM is encouraged to use humor where appropriate, ensuring engaging and emotionally aligned responses. Responses are concise (under 25 characters) and based on real-time game context to enhance relevance and the player's experience.

5 Study 1: Evaluation of Acoustic-Based Emotion Estimation in Games

To validate our ultrasonic emotion detection, we invited the 24 participants from our formative study to a follow-up in-lab experiment. Twenty individuals (13 males, seven females; aged 18- 35) accepted the invitation, while four declined. Real-time interventions can alter players' natural emotional trajectories and suppress the strong negative affect required for reliable ground-truth labeling. To avoid such confounds and ensure unbiased valence-arousal measurements, Study 1 used an offline version of the system without interventions. After briefing each participant on the study's procedures and obtaining their informed consent, we collected a total of 35 hours of audio data, including 7 hours of emotional expressions captured during gameplay.

5.1 Procedure

Game Types. To elicit a diverse range of emotional responses within a limited timeframe, participants played three different types of games: Fast-pace Game (MOBA-style): A fast-paced multiplayer online battle arena (MOBA) game was selected due to its highly competitive nature, where win/loss outcomes significantly influence emotions. Moderate-pace Game (Strategy Simulation): A moderate-paced strategy simulation game was chosen to create a more thoughtful, planning-oriented experience. While it lacks the immediate intensity of action games, it still evokes emotional responses through the tension of managing resources, building strategies, and overcoming challenges. High-intensity Game (Battle Royale): A battle royale game was included to provide moments of high emotional engagement. With its unpredictable pacing, players experience adrenaline rushes during combat and heightened emotional states as the match progresses towards its final stages, fostering a sense of anxiety and excitement.

Experiment Procedure. Each participant completed 5 gaming sessions with a total duration of approximately 90–120 minutes, each lasting 15–25 minutes with breaks scheduled in between. During the experiments, participants' facial expressions were recorded using the smartphone's front camera, and these videos served as the ground truth source for ultrasonic spectrogram inputs.A preliminary version of the RageSense application ran continuously in the background while participants were playing the game. The app recorded IMU data and ultrasonic-derived features.

Annotation Process and Reliability. Building on the established principle that individuals are the most reliable annotators of their own emotions [59, 80], we adopted a self-annotation methodology. We invited the same 20 participants from our study to label their own facial expression videos, which were recorded using the smartphone's front camera during gameplay. To prepare them for this task, all participants completed a systematic training program structured in three stages: (1) theoretical instruction on the valence-arousal dimensional emotion model; (2) guided practice, where they learned to map valence-arousal scores to example video clips with expert reference ratings; and (3) a calibration phase, where they performed trial annotations with immediate feedback until their ratings achieved a consistent and reliable level.

We employed a 9-point Likert scale for valence-arousal annotation, a method widely adopted in prior studies [26, 42, 51]. Videos were sampled at 3 fps, and each frame was rated by 10 randomly selected participant-annotators, with the final value determined through majority voting. Ratings were then linearly normalized from $[-4, 4]$ to $[-1, 1]$. The ground truth for each ultrasonic spectrogram was obtained by averaging the valence-arousal labels of its time-aligned video frames.

Annotation reliability was evaluated using Krippendorff's α, yielding $\alpha = 0.689$ for valence and $\alpha = 0.661$ for arousal. According to Krippendorff's guidelines, valence annotations are acceptable for exploratory studies, while arousal shows slightly lower agreement but remains consistent with existing valence-arousal datasets (typically $\alpha = 0.4$–0.65). [9, 61].

5.2 Metric

Study 1 aims to assess the accuracy of the acoustic-based emotion detection algorithm. To evaluate the performance, we calculate the error between the estimated valence and arousal values and the ground truth, which is derived from the facial expression analysis of the recorded video. The following evaluation metrics are employed:

Table 1: Performance comparison of different models on valence-arousal regression.

Model	CCC_V	CCC_A	PVA
Video based[58]	**0.6411**	**0.6356**	**0.6384**
MobileNet	0.4430	0.4438	0.4434
MobileNet + LSTM	0.4675	0.4521	0.4598
Transformer	0.4821	0.4939	0.4880
Transformer+ LSTM	0.5771	0.5542	0.5657
Sparse Transformer+ LSTM (Ours)	0.6115	0.6215	0.6165

Root Mean Square Error (RMSE). This metric quantifies the difference between the predicted and actual values. It provides an understanding of the magnitude of errors, where lower values indicate better accuracy.

Concordance Correlation Coefficient (CCC_V/CCC_A)[58]. This metric evaluates the agreement between predicted and ground truth values by simultaneously assessing both their precision and accuracy. It is particularly valuable for emotion regression tasks, as it reflects how well the model captures not only the linear correlation but also the absolute closeness of the predictions to the actual values, thus serving as a comprehensive indicator of predictive reliability.

Proportion of Variance Accounted (PVA)[58]. This metric is defined as the mean of the concordance correlation coefficients for valence (CCC_V) and arousal (CCC_A), expressed as PVA = (CCC_V + CCC_A) / 2. It provides a balanced evaluation of the model's predictive performance across both emotional dimensions, and is particularly useful for quantifying how effectively the algorithm captures the underlying patterns of affective dynamics.

5.3 Valence and Arousal Estimation Performance

In this section, we evaluate and present the accuracy of the user-independent model and impact of different model architectures.

5.3.1 Leave-one-subject-out Cross Validation. Maintaining consistent performance across different users has been a long-standing challenge for most ubiquitous sensing systems. As shown in Figure 6, we employed a *leave-one-subject-out* cross-validation strategy to evaluate the correlation coefficients (CCC) for valence (CCC_V), arousal (CCC_A), and the combined PVA (Pearson's Variance Average) metrics. The results show that while individual subjects exhibit varying performance, the system consistently achieves a high correlation across subjects, with most subjects demonstrating CCC values above 0.5. Notably, the average performance for CCC_V and CCC_A remained consistently strong, suggesting that the system is robust and adaptable to different users.

A deeper analysis of the outlier cases, highlighted in Figure 6, reveals two core challenges impacting system performance. The first is a physical challenge at the signal level. For P1 and P16, intense gameplay engagement resulted in significant head movements, introducing substantial motion artifacts. These artifacts corrupted the facial signal, obscuring subtle expressions and undermining the integrity of the feature extraction process, which in turn degraded prediction accuracy. In contrast to this data-level challenge, P9's case exposes the problem of subjectivity in emotion annotation. The participant consistently maintained a serious expression with minimal affective cues. However, this objective "poker face" was subjectively misinterpreted by human annotators as a negative emotion (e.g., frustration), introducing a systemic bias into the ground-truth labels. Consequently, although our model correctly captured the absence of facial activity, this discrepancy with the flawed ground truth artificially deflated the final correlation scores. Together, these cases highlight a dual dilemma for models in naturalistic settings: they must achieve physical robustness against signal-level noise while also depending on semantic accuracy at the label-level for effective learning.

These results suggest that, despite variations in individual player behavior, the system demonstrates robustness and adaptability across different users, confirming the model's effectiveness for practical applications.

5.3.2 Leave-One-Game-Out Cross-Validation. To evaluate the generalization capability of our system across different game scenarios, we employed a *leave-one-game-out* cross-validation strategy on a per-participant basis. For each of the five games played by our 20 participants, we iteratively designated one game as the test set while using the remaining four for training. As illustrated in Figure 6, the results underscore the system's high performance, with the CCC_V, CCC_A, and PVA metrics generally exceeding 0.6 across all game scenarios.

These findings provide strong evidence that the model maintains robust and reliable predictive power across varied gaming contexts. This consistent and strong performance highlights the system's adaptability and signals its significant potential for practical applications in diverse interactive environments.

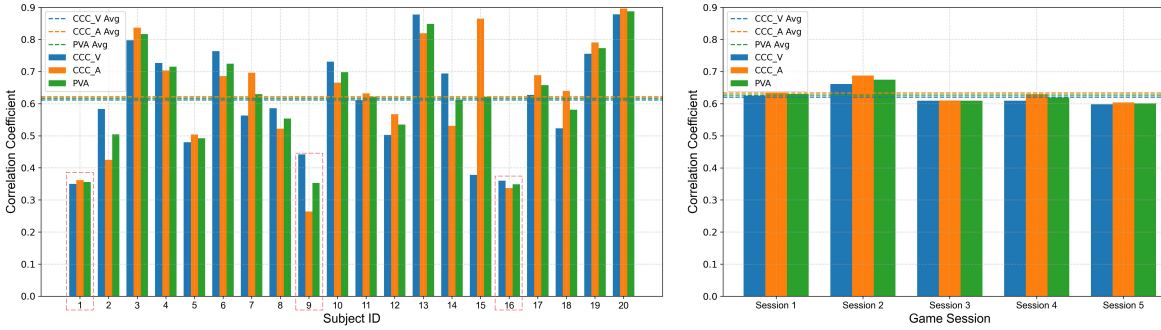
5.3.3 Impact of model architecture. To better map spectrogram features to valence and arousal values, we compared several network architectures, including CNN-based models like MobileNet, transformer-based models, and hybrid models like MobileNet + LSTM and Transformer + LSTM. We also implemented an EfficientNet model using facial images for comparison with state-of-the-art visual emotion recognition methods.

As shown in Table 1, our Sparse Transformer + LSTM model outperforms all other models, achieving CCC scores of 0.6115 for valence, 0.6215 for arousal, and a PVA of 0.6165. The Transformer + LSTM hybrid also showed strong performance, with CCC scores of 0.5771 for valence, 0.5542 for arousal, and a PVA of 0.5657. These results emphasize the benefit of combining transformer encoders for feature extraction and LSTM decoders for temporal modeling.

Overall, these results confirm that the architecture choice is crucial for emotion estimation, with our Sparse Transformer + LSTM model effectively capturing subtle emotional changes and performing almost as well as video-based SOTA methods[58].

5.4 Hand Operation Segmentation Efficiency

To validate the effectiveness of our proposed hand operation segmentation module, we collected 8 gameplay sessions (4 MOBA and 4 Simulation Strategy), totaling 135 minutes, specifically capturing



(a)Leave-one-subject-out cross validation results for 20 participants. (b)Leave-One-Game-Session-Out Cross-Validation Results.

Figure 6: Comparison of Leave-One-Out Cross-Validation Results Across Participants and Games

Table 2: Segmentation and detection statistics (Mean and SD) across two game types.

Metric	MOBA		Simulation Strategy	
	Mean	SD	Mean	SD
Total gameplay duration(s)	986.36	±128.66	1022.89	±185.99
Detection time ratio(%)	32.63	±4.89	4.05	±2.87
Detection accuracy(%)	85.06	±7.92	90.35	±8.50

high-intensity periods such as team battles and rapid gestures. To establish a ground truth, two annotators independently reviewed and manually labeled these sessions based on a clear set of criteria for high-intensity operations: (1) Operation Frequency, defined as continuous rapid clicks at a rate of 2 times per second or more; (2) Hand Movement Intensity, including actions like forceful taps and broad screen swipes; and (3) Action Duration, requiring the intense operation to be sustained for over 2 seconds.

We then evaluated the module's performance by comparing the IMU-based automatic detection windows against the manually annotated ground truth. As detailed in Table 2, our module performed robustly across both game genres, achieving high detection accuracies of 85.06% for MOBA and 90.35% for Simulation Strategy, which confirms its general applicability. The slightly higher accuracy in Simulation Strategy games may be attributed to their more predictable interaction patterns, whereas the sudden, chaotic bursts of action in MOBA team fights present a more complex segmentation challenge. A more insightful finding emerged from the proportion of gameplay time identified as high-intensity (Detection Time Ratio): the average ratio for MOBA games was 32.63%, substantially higher than the 4.05% for Simulation Strategy games. This stark contrast provides quantitative evidence for the differing operational demands of the genres, confirming that MOBA gameplay requires far more frequent and intense hand interactions. In summary, these results demonstrate that our module accurately identifies and filters out noise from high-intensity operations, providing a crucial foundation for the accuracy and stability of the downstream emotion detection system by ensuring sensing occurs only when the device is relatively stable.

5.5 Robustness Evaluation

To validate the robustness of RageSense in real-world application scenarios, we recruited 5 participants out of 20 participants for the

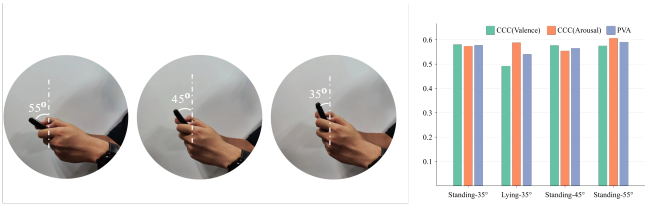


Figure 7: Left: An example of holding the smartphone with a specific orientation. Right: averaged similarity metrics of VAs over different holding angles.

robustness study, with each completing 10 sessions. To ensure fair model training and evaluation, the collected data were balanced across different emotional states. We then conducted performance tests under varying holding angles and body postures.

5.5.1 Impact of Holding Angles. Different from PC games, mobile games need the user to hold the smartphone all the time. Through a group survey, we found that 35-55° is the most common and comfortable angle range to hold the smartphone horizontally, such that the line of sight is perpendicular to the smartphone screen. However, different users may have various preferences for smartphone-holding angles during mobile gameplay, as shown in Fig. 7, which includes three smartphone angles (35°, 45°, and 55°) facing the user's face. Therefore, we implemented the addition experiment (N=5) to examine how the smartphone orientation would affect the estimation performance. During the registration phase for training, the subjects recorded the echoes with the smartphone held at a default angle (θ= 45°). We then captured the reflected echoes with different smartphone angles from 35° to 55° to simulate the varying smartphone holding behaviors in daily gameplay.

From Fig. 7, it is observed that the accuracy is highest at a 55° holding angle, while the accuracy for the 35° and 45° angles is similar and slightly lower. This result is contrary to our initial expectations of a steady decline in accuracy with increasing angle. The higher accuracy at 55° may be due to better alignment of the face with the smartphone microphone's coverage area compared to the 35° and 45° angles, where the face region is more displaced.

5.5.2 Impact of Body Postures. As the sound wave intensity decreases in inverse proportion to the squared distance between two objects, as well as due to multi-path channel fading, the reflected

signal quality may vary depending on the specific locations. In previous evaluations with the default experimental setting, we asked participants to sit and keep the smartphone 20 cm away from the face. To investigate the impacts of different user body postures on recognition performance, we implemented another robustness tests for different postures. In the experiment, we asked five participants to perform the experiments in two common postures, including sitting on a chair, and reclining on the bed.

The experimental results reveal that body posture significantly influences system performance. Specifically, the sitting posture yields superior performance across all metrics, achieving a concordance correlation coefficient (CCC) of 0.611 for valence, 0.621 for arousal, and a combined PVA score of 0.616. In contrast, the reclining posture leads to noticeable degradation, with CCC values of 0.492 (valence), 0.588 (arousal), and a PVA score of 0.540. This performance drop may be attributed to less stable hand-holding behavior and increased variability in the relative positioning between the smartphone and the user's face when reclining. Unlike the seated posture—where users often stabilize their arms against a table, maintaining a consistent and optimal sensing distance—the reclining position may introduce subtle posture shifts due to gravitational effects on the arm, leading to degraded signal quality and recognition accuracy. These findings suggest that user posture plays a non-trivial role in wireless sensing applications and must be considered when designing robust and generalizable affect recognition systems.

5.5.3 Robustness to Game Audio. A key consideration is the system's robustness to the game's own audio, as both the ultrasonic probe signal (18–22 kHz) and the audible game sounds (<10 kHz) are emitted simultaneously from the same speaker. Our design leverages this clear spectral separation. We empirically validated this by measuring the Signal-to-Noise Ratio (SNR) of the received echo, observing only a negligible 0.34 dB drop when game audio was active. This confirms that concurrent game sound has virtually no impact on sensing performance.

5.5.4 Robustness to Dynamic, Noisy Environments, and User Characteristics. We expanded the robustness evaluation to include dynamic and noisy environments, as well as various user characteristics.The results are summarized in Table 3. We invited 5 participants from Study 1 to participate in a supplementary session for environmental robustness validation.

Noisy Environment. We evaluated RageSense in a high-noise indoor public environment (i.e. cateteria, 78.42 dB) characterized by strong non-stationary ambient noise (e.g., human chatter, utensil impacts, and broadcast noise). Although this noise is outside the ultrasonic band, its harmonic components partially overlap with the sidelobes of the transmitted near-ultrasonic signal, weakening the quality of the received echoes. As a result, the system showed a mild performance drop (CCC_V ↓8.81%, CCC_A ↓8.08%), but the accuracy remained within an acceptable range, indicating that the core ultrasonic features were still preserved despite heavy acoustic interference.

Dynamic Environment. we conducted tests in an outdoor walkway with frequent pedestrian movement and varying ambient noise levels (70.11 dB). This environment contained transient occlusions and secondary reflections from moving bodies, which could potentially disturb the non-line-of-sight echo path. However, because the primary acoustic path—between the smartphone and the user's face—is substantially stronger than reflections from surrounding motion, performance degradation remained limited (CCC_V ↓2.39%, CCC_A ↓4.92%). These results demonstrate that RageSense maintains stable accuracy even under typical everyday human movement.

Skin Tones. We included participants with medium-to-dark skin tones (N=3). Compared to lighter-skin participants in the controlled study, we observed a moderate decrease in valence-arousal estimation concordance ($CCCV$ ↓9.03%, $CCCA$ ↓14.35%). While we did not collect sufficient data to identify the underlying cause, we note that two of these participants had substantial facial hair, which may partially obscure fine-grained geometric changes captured through ultrasonic reflections. We note that the sample imbalance reflects our local demographic pool.

Accessories and Hair Styles. We examined robustness to common user attributes, including eyewear and hair configuration. Performance remained consistent across users with glasses (N=13) and without glasses (N=7), indicating that minor multipath reflections from eyeglass frames had limited impact. Similarly, both long-hair (N=9) and short-hair (N=11) participants showed negligible performance differences (PVA 0.6118–0.6221). These results suggest that typical variations in accessories and hair styles exert minimal influence on the primary smartphone-to-face echo path.

These robustness experiments, while based on modest sample sizes, indicate that RageSense maintains reasonable performance across noisy/dynamic environments and diverse user characteristics, supporting its applicability in real-world mobile gaming scenarios.

5.6 Discussion

The results demonstrate the significant effectiveness of acoustic-based emotion detection in gaming scenarios. First, the system performed exceptionally well in tracking emotional fluctuations within a single gaming session, proving the feasibility of ultrasonic-based emotion detection in dynamic gaming environments. We chose to allow the system to learn across different games to adapt to changes in user emotions and external disturbances. This approach enables the system to maintain stable performance across multiple sessions, and despite some fluctuations, it still effectively captures emotional changes.

However, although the system excels at filtering hand movements, variations in smartphone holding angles can still lead to inaccurate predictions, highlighting the challenges of emotion detection in real-world environments. Despite the influence of hand posture on detection, the system was still able to effectively identify intense emotional fluctuations, as both valence and arousal are time-varying. To further improve the system's robustness and practicality, the subsequent Study 2 will explore how to enhance emotion detection accuracy through large models and intervention systems that incorporate game content.

Table 3: Robustness evaluation across noisy and dynamic environments and diverse user characteristics.

Condition	CCC_V (Mean ± SD)	CCC_A(Mean ± SD)	PVA (Mean ± SD)
High-Noise Environment	0.5576 ± 0.0154	0.5713 ± 0.0137	0.5644 ± 0.0141
Dynamic Environment	0.5969 ± 0.0006	0.5909 ± 0.0041	0.5939 ± 0.0018
Skin Tone (Medium-Dark)	0.5563 ± 0.0109	0.5323 ± 0.0000	0.5443 ± 0.0025
Eyewear (with Glasses)	0.6232 ± 0.0296	0.6118 ± 0.0298	0.6175 ± 0.0272
Eyewear (without Glasses)	0.5841 ± 0.0198	0.6441 ± 0.0334	0.6141 ± 0.0150
Hair (Long Hair)	0.5974 ± 0.0259	0.6468 ± 0.0226	0.6221 ± 0.0170
Hair (Short Hair)	0.6230 ± 0.0251	0.6007 ± 0.0375	0.6118 ± 0.0272

6 Real-time System Implementation

Study 1 showed the effectiveness of the proposed approach in estimating user's emotion status, which is the foundation of emotion intervention.

6.1 Prototype Application

To assess the practicality of real-time emotion estimation and explore the effectiveness of in-game interventions, we developed a prototype mobile application for the Android OS (Appendix A). The app operates in the background, emitting inaudible ultrasound waves and analyzing the characteristics of the reflected echoes to predict the user's emotional state, including valence and arousal levels. When significant fluctuations in these emotional parameters are detected, the app captures a screenshot of the current game content and sends it, along with historical valence and arousal data, to a cloud server. The cloud server then uses a large language model (LLM) API to interpret both the game content and the user's emotional state, generating a personalized intervention message that is sent back to the mobile app. This message is displayed as a brief, semi-transparent pop-up, which is strategically positioned away from core gameplay controls and interactive elements. It automatically disappears after 5 seconds to avoid interfering with the fast-paced nature of gameplay. This ensures that users receive targeted emotional regulation suggestions without disrupting the intensity or flow of their game.

To benchmark the performance of our emotion detection system, we also created two baseline applications: one that triggers interventions based on predefined contextual cues, and another that triggers interventions at random intervals. This comparison allows us to evaluate the relative effectiveness of context-aware interventions versus arbitrary ones.

6.2 Runtime Performance

To evaluate the computational overhead of detecting rage reactions using the RageSense app, we analyzed system latency, as summarized in Table 4. This step involves various signal processing tasks, with the Vivo X90 requiring 133 ms, the Redmi K60 requiring 175 ms, and the OnePlus Ace3 requiring 162 ms for feature generation. Following optimization, the feature generation step involves various signal processing tasks, requiring only 133 ms on the Vivo X90, 175 ms on the Redmi K60, and 162 ms on the OnePlus Ace3. The subsequent valence-arousal estimation step runs faster than the intervention generation, with latencies of 295 ms, 248 ms, and 202 ms on the respective devices.

Table 4: The system latency on different smartphones. The latency results were obtained using a 5-second audio sample averaged over 10 runs. The intervention generation latency depends on the connection speed to the LLM in the cloud server.

Latency (ms)	Vivo X90	Redmi K60	Oneplus Ace3
Feature Gen.	133	175	162
VA Estimation	295	248	202
Intervention Gen.	500	500	500
Total	928	923	864

The system enables continuous emotion detection during gameplay. Despite network latency and LLM processing delays, these factors remain manageable and do not lead to significant accumulation. The total system latency, ranging from 864 ms on the OnePlus Ace3 to 928 ms on the Vivo X90, is well within the 2-second striding window, ensuring real-time feedback. This enhanced performance demonstrates the system's ability for real-time emotion detection, making it suitable for interactive gaming applications.

To quantify the power overhead introduced by RageSense, we measured battery drain on a Vivo X90 (4810 mAh) under two conditions: Game Only and Game + RageSense. Each session lasted 30 minutes and repeated 5 times with fixed screen brightness (50%) and network conditions. Battery Historian was used to obtain per-component power share. Game Only: total drain = 6.08%, Game + RageSense: total drain = 7.68%. RageSense therefore introduces an additional 1.60% absolute battery consumption. This overhead is small relative to the dominant contributors—the game engine (58%) and the screen (18%)—with RageSense accounting for only 24% of the total consumption during the combined condition. Memory profiling shows that RageSense occupies approximately 330 MB total PSS (Proportional Set Size), which is well below the 1.8 GB used by the MOBA game and remained stable without leakage across all runs.

6.3 Impact of Smartphone Models

Theoretically, most high-end smartphone speakers are designed to produce sound within the frequency range of 100 Hz to 20 kHz, which is capable of reproducing near-ultrasound frequencies up to 19 kHz. However, given the variation in hardware specifications across models, it is important to investigate the quality of the near-ultrasound signal used in the RageSense system on different smartphone devices. In this section, we assess the signal quality loss of the spectrogram generated from near-ultrasound signals.

We measured the Signal-to-Noise Ratio (SNR) as the metric across four Android smartphones: Redmi K60, Huawei Mate 30, Vivo X90, and Samsung S9. Our results show that the near-ultrasound signal quality across these models is relatively consistent, with SNR values ranging from 15.87 dB to 16.41 dB.

7 Study 2: In-field Study with real-time Intervention In Mobile Gaming

To evaluate the effectiveness of RageSense's emotion-aware, LLM-based intervention system, we conducted a field study with 53 new participants. Given the relatively small sample size, we utilized a Latin Square design to ensure each participant experienced all three intervention models. The order of conditions was counterbalanced across participants to mitigate potential sequence effects.

7.1 Intervention Conditions

Throughout the gameplay, RageSense's emotion detection module continuously monitored participants' emotional states. The key distinction between the intervention models was the nature of the intervention triggered when a participant's negative emotional indicators exceeded a predefined threshold:

Control condition (Pre-defined Text-Based Intervention). : In this condition, intervention requests were triggered based on predicted valence-arousal values derived from the emotion detection model. The system selected intervention messages randomly from a predefined text library, offering non-adaptive guidance. The messages were delivered in the participant's preferred format (either text or voice).

Proposed condition (LLM-based Personalized Intervention). : This condition also used predicted valence-arousal values from the emotion detection model to trigger interventions. However, unlike the Control Condition, the intervention messages were generated dynamically by a large language model (LLM). The system processed real-time game screenshots and emotional data to generate personalized, contextually relevant suggestions or supportive messages. The intervention was adjusted dynamically based on the participant's emotional state, aiming to provide more tailored and context-aware interventions.

Baseline condition (Random Timing-based Intervention). : In this condition, interventions were triggered at random intervals, irrespective of the participant's emotional state. The intervention messages were randomly selected from the predefined text library, offering non-adaptive and uncontextualized guidance. This condition served as the baseline for comparison.

7.2 Metric

The field study spanned over three weeks, with each participant completing multiple sessions of gameplay. As the study took place in the field, the duration of each session varied depending on the participant's schedule, but it was required that each session lasted at least one hour. To evaluate the effectiveness of the interventions, two primary types of data were collected:

Emotional Change Metrics: The primary measure was the immediate impact of interventions on players' emotional states. With user consent, we accessed users' valence-arousal values during their interaction with the app, along with in-game recording data, to assess the effectiveness of the intervention. Emotion data from one minute before and one minute after each intervention was extracted and changes in valence (pleasantness) and arousal (tension level) were computed. The effectiveness of the intervention was assessed by comparing the average post-intervention emotional improvement (increase in valence and decrease in arousal) for each participant across all intervention instances.

Subjective Questionnaire Responses: Participants completed a post-experiment survey assessing their perceptions of the intervention. The survey included satisfaction ratings (on a five-point Likert scale), the perceived helpfulness and contextual relevance of the intervention, and general acceptability (e.g., whether they found the interventions intrusive). The experimental group provided additional feedback on the LLM-generated messages, including their perceived naturalness and empathetic quality, while the control group was asked about their impressions of the pre-defined messages.

7.3 Procedure and Participants

The study was conducted over a three-week period, with participants completing one session per week. A call for new participants who had not been involved in Study 1 was posted in the university community, where participants were selected based on the following criteria: they must own an Android smartphone and have prior experience with mobile gaming. Consistent with the demographic distribution of MOBA players—where young adults (18–30 years old) constitute the dominant player segment according to prior market reports[2, 3] and game analytics[20]. Our recruitment focused on university students. A total of 85 participants responded to the call, from which 15 participants discontinued their participation, 10 encountered software compatibility issues, and 7 raised privacy concerns. In the end, 53 participants completed the study and provided feedback.

To ensure the robustness of the results, we employed a Latin Square design to mitigate order effects. Each participant was randomly assigned to a specific order of intervention conditions (Control, Baseline, or Proposed), with each intervention being experienced in a different week. The order of interventions was counterbalanced across participants, ensuring that each intervention model was tested evenly throughout the study period. Participants were free to schedule their gameplay sessions at their convenience, ensuring a more naturalistic and flexible in-field testing environment. The study has been approved by the Internal Review Board of [the institute name is hidden for double-blinded review].

7.4 Quantitative Analysis

7.4.1 Effectiveness of Intervention Time. The effectiveness of the intervention was evaluated based on changes in Valence Increase and Arousal Drop across different intervention conditions: Control (valence-arousal triggered), Personalized (valence-arousal-triggered), and Baseline (randomly). As shown in the Figure 8, the Personalized intervention (proposed) resulted in a significant increase in valence and a greater arousal drop compared to both the Control and Baseline conditions in Simulation Strategy games. Specifically, a marked

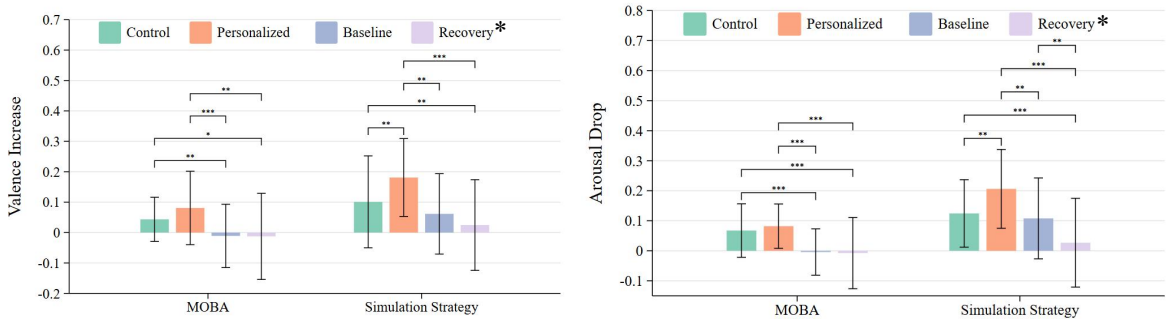


Figure 8: Calculated valence-arousal changes across three intervention methods and Recovery condition shown as a reference for naturally occurring recovery. The Recovery data are derived from the 50% trigger-probability study.

valence increase was observed in the Personalized condition, highlighting the positive emotional impact of in-game interventions. These results suggest that during-game interventions (Personalized) have a more significant effect in games like Simulation Strategy, where emotional states are more stable and decisions can greatly influence the outcome of the game.

To contextualize these findings, we further included a Recovery reference condition derived from a subset of participants in the in-field study (N=10 out of 53), based on the 50% trigger-probability design, where interventions were randomly omitted in half of the eligible moments[5, 8]. This provides an estimate of naturally occurring affective recovery within the same session and subject. As shown in Fig. 8, the magnitude of valence and arousal increase in the Recovery moments was substantially smaller than in any intervention method, confirming that the observed improvements are attributable to the interventions rather than spontaneous emotional recovery.

7.4.2 Effectiveness of Content Understanding. The effectiveness of Content Understanding was assessed by analyzing changes in valence and arousal as a result of intervention content. As shown in the Figure 8, Personalized interventions significantly improved valence in both game types, with stronger effects observed in Simulation Strategy games compared to MOBA games. In particular, the Personalized intervention resulted in a clear valence increase compared to the Control condition, which remained relatively stable across both game types.

These findings suggest that content-specific interventions are most effective in games like Simulation Strategy, where emotional responses are stronger due to slower-paced decision-making and more significant gameplay outcomes. In MOBA games, the rapid pace of the game and team dynamics may limit the emotional impact of interventions.

7.4.3 Behavioral Measures. We used normalized playtime as a behavioral measure to evaluate the intervention effects. Playtime is a key indicator of player engagement and emotional regulation. We hypothesize that the intervention would affect playtime by improving emotional management, particularly in games where individual performance plays a larger role.

The figure 9 shows normalized playtime for the Simulation Strategy and MOBA strategies across three conditions: Control, Personalized, and Baseline. In Simulation Strategy, the personalized intervention had the highest playtime, with the control condition

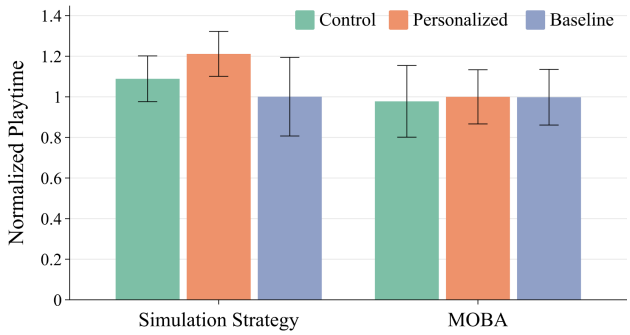


Figure 9: Playtime changes comparison across three intervention methods.

showing a slight decrease, and the baseline showing the lowest. For MOBA, there was little difference between the control and personalized conditions, while both were higher than the baseline. These results suggest that the intervention is more effective in Simulation Strategy games. For MOBA games, which are team-based, players are less likely to quit mid-game even under negative emotions, leading to little impact on average playtime. In contrast, Simulation Strategy games are more individually driven, making players more susceptible to emotional influences, and thus, more likely to be affected by the intervention.

7.5 Subjective User Feedback

In addition to the quantitative analysis of intervention effectiveness, participants' post-survey responses and comments provided valueable insights.

7.5.1 Comparative Effectiveness of Intervention Strategies. Overall, participants had a clear preference for the personalized (ours) intervention method over the control and baseline methods. As shown in Figure 10, across all evaluated aspects, the personalized intervention consistently outperformed the baseline with statistical significance.

For *Experience Enhancement*, the personalized approach achieved an average score of 4.5 (SD = 0.58), significantly higher than both the control (M = 3.1, SD = 0.36, $p < .0001$) and baseline methods (M = 2.9, SD = 0.48, $p < .0001$). Similarly, in *Regulation Effectiveness*, our method reached 4.2 (SD=0.48) , exceeding the control (M = 3.5, SD=0.78, $p < .05$) and baseline (M = 3.0, SD=1.3, $p < .01$).

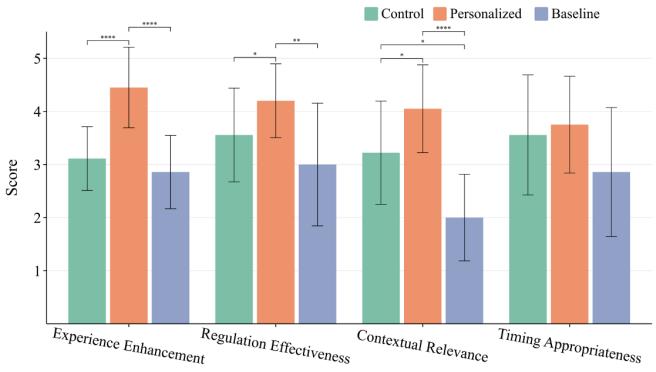


Figure 10: Survey feedback across four aspects of the intervention.

The advantage of personalization was also evident in *Contextual Relevance*, where our system scored 4.0 (SD = 0.68), outperforming both control (M = 3.2, SD = 0.94, $p < .05$) and baseline (M = 2.0, SD = 0.67, $p < .0001$). Lastly, for *Timing Appropriateness*, our intervention (M = 3.7, SD=0.83) outpaced the baseline (M = 2.9, SD= 1.48, $p < .05$), though its advantage over the control condition (M = 3.5, SD=1.28) did not reach statistical significance.

Furthermore, to directly evaluate the potential for gameplay interruption—a key usability concern—the survey included specific questions on the intervention's disruption and intrusiveness. Participants rated the personalized intervention highly for its non-disruptive nature (M=4.23, SD=0.42) and non-intrusiveness (M = 4.46, SD = 0.50). These scores were not only high in absolute terms, with 87% of participants providing a rating of 4 or higher on a 5-point scale, but also significantly outperformed the control and baseline conditions. This feedback confirms that our design approach successfully delivered timely emotional support without negatively impacting the core user experience.

7.5.2 Light and Heavy Gamers Report Higher Satisfaction Across Dimensions. We further analyzed participants by their self-reported daily game playtime, categorized into light, moderate, and heavy gamers. As illustrated in Figure 11 (left), light gamers consistently reported higher satisfaction across all evaluated dimensions. In particular, they gave the highest ratings for *Experience Enhancement* (M = 4.6, SD = 0.7), compared to moderate gamers (M = 4.2, SD = 0.8) and heavy gamers (M = 4.3, SD = 0.9). Similarly, light gamers rated *Regulation Effectiveness* notably higher (M = 4.5, SD = 0.8), with a significant difference observed between light and heavy gamers ($p < .05$).

Although no other dimensions reached statistical significance, light gamers consistently rated higher in *Contextual Relevance* and *Timing Appropriateness* as well, suggesting that less frequent gamers may perceive greater value in personalized emotional interventions. This trend indicates that light gamers, who might be less accustomed to intense gaming sessions, particularly appreciate the supportive feedback provided by our system.

7.5.3 Emotionally Sensitive Users Benefit More from Interventions. We also analyzed user responses based on emotional sensitivity, categorizing participants into two groups: *Emotionally Sensitive* and *Emotionally Stable*. This classification was based on our custom Emotion Sensitivity Index (ESI), derived from a pre-study questionnaire. The ESI is the sum of scores from three items: 1) the number of negative in-game scenarios experienced in the past (0–7), 2) the frequency of negative emotions in the last month (a 4-point scale), and 3) the typical time it takes to feel frustrated during gameplay (a 4-point scale). We applied a median split to the total ESI scores (Mdn = 8, range: 2–15), classifying participants with scores at or above the median as *Emotionally Sensitive*, and the remainder as *Emotionally Stable*.

As shown in Figure 11 (right), emotionally sensitive users consistently rated the system more favorably across all metrics. The difference was statistically significant for *Emotional Understanding*, where sensitive users gave higher scores (M = 4.6, SD = 0.8) compared to stable users (M = 4.1, SD = 0.9), with $p < .05$.

Although differences in *Regulation Effectiveness* and *Experience Enhancement* did not reach significance, the sensitive group consistently rated higher across these aspects as well. These findings suggest that emotionally sensitive users, who are more likely to experience fluctuating emotional states during gameplay, benefit more from the system's personalized interventions.

7.6 Case Study: Regulating Emotion in a Losing Game

To provide a compelling qualitative demonstration of RageSense's in-the-wild effectiveness, we present an in-depth case study. The participant, recruited from our study 1, is a high-frequency player who games 3-5 hours daily and reported experiencing daily negative emotions from gameplay. This profile makes them an ideal representative of our target demographic: players whose well-being is significantly impacted by emotional dysregulation. We analyze this player's emotional trajectory during a challenging competitive match, illustrating how our system's targeted interventions helped them regulate their emotions despite a significant in-game disadvantage (visualized in Figure 12).

7.6.1 Initial Setbacks and Emotional Distress. The match began unfavorably. As shown in the Gold Difference Timeline (top panel), frequent deaths and early-game ganks placed the player's team at a significant economic disadvantage. This adverse situation manifested directly in their emotional state (bottom panel). Their Valence (blue line) plummeted into the negative spectrum, indicating sustained feelings of displeasure and frustration. Concurrently, their Arousal (red line) exhibited sharp, erratic spikes corresponding to in-game setbacks, painting a clear picture of a player entering a negative emotional spiral.

7.6.2 Targeted Interventions and Emotional Trajectory. RageSense delivered three interventions at moments of detected acute distress, marked by yellow stars. Each intervention had a distinct and observable impact on the player's emotional trajectory.

First Intervention (approx. 1:00): Stabilizing an Early-Game Slump. Following an initial setback, the player's Valence hit a low point. The intervention at 1:00 appears to have curtailed this downward trend. We observe an immediate, albeit modest, rise in Valence towards a more neutral state, while the preceding Arousal spike

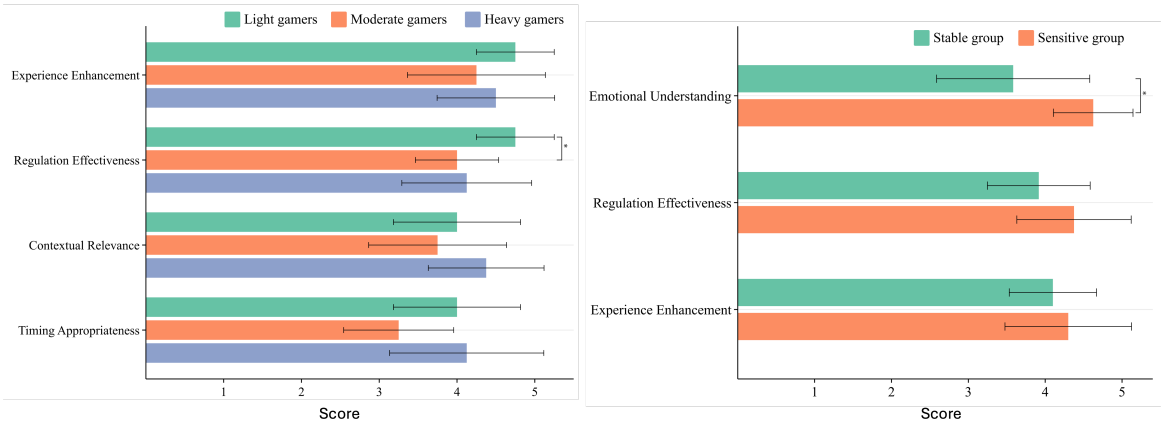


Figure 11: Left: Survey feedback of Experience Enhancement, Regulation Effectiveness, Contextual Relevance and Timing Appropriateness across three gamer levels. Right: Survey feedback across two emotional sensitivity groups.

subsides. This suggests the intervention successfully provided a moment of emotional stabilization, preventing further decline.

Second Intervention (approx. 4:00): Fostering Resilience in the Mid-Game. As the gold disadvantage deepened around the 4:00 mark, the player experienced another significant emotional dip. The second intervention was triggered in response. Post-intervention, the Valence curve shows a more pronounced and sustained upward trend, demonstrating an improved emotional state despite the ongoing in-game challenge. The intervention seemingly helped the player build emotional resilience.

Third Intervention (approx. 8:30): Maintaining Composure Under Pressure. During a late-game event, the player's Arousal spiked sharply, indicating a moment of high stress. The third intervention was deployed, and we can observe its effect in how rapidly this Arousal peak was reduced. Crucially, the player's Valence remained relatively stable instead of crashing. This suggests the intervention was effective in helping the player maintain composure during a critical, high-pressure situation.

7.6.3 The Cumulative Effect: A Player's Journey from Frustration to Focus. The cumulative impact of these interventions is best illustrated by the player's own post-game reflection:

"I was frustrated with my performance after dying during the early laning phase(approx. 4:00). The system's intervention soothed my frustration and, crucially, reduced the weight of my self-blame by reminding me that one mistake wouldn't decide the entire match's outcome. Later, in an important team fight where I was the initiator, my key abilities failed to connect, leading directly to our team's defeat and a significant gold gap(approx. 8:00). I felt the loss was entirely my fault. The system then prompted me to recognize that, ultimately, it is just a game. This allowed me to consciously calm down and refocus my emotions. This internal process of self-regulation was absolutely critical. With a renewed focus, we won a decisive late-game team fight, started to reclaim the economic advantage, and ultimately, secured the victory."

This player's Statement Illustrates our quantitative data in a personal experience, illustrating how the interventions facilitated a crucial shift from frustration to resilient focus, ultimately contributing to their success.

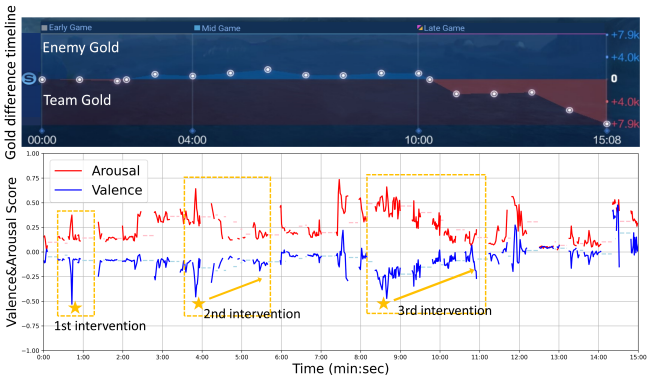


Figure 12: A case study of game progression and emotional response with interventions.

8 Discussion and Limitation

8.1 Beyond the Individual:The Social Context of Emotion

A key limitation of RageSense is its current focus on individual emotion, which may not fully capture the rich social context of multiplayer gaming[44]. This can occasionally lead to interventions that are contextually inappropriate. A promising direction for future work is to extend the system to foster interpersonal emotion regulation (iER) skills, which are vital for teamwork. We envision evolving RageSense from a reactive "regulator" to a proactive social-emotional learning (SEL) "partner". Such a system would analyze team-level emotional data to facilitate supportive interactions, for instance, by suggesting an encouraging message when a teammate is struggling. This is particularly valuable in common scenarios like friends playing together, shifting the goal from merely mitigating individual rage to actively fostering more cohesive and positive team dynamics, ultimately enhancing the shared social experience

8.2 Not All Game Contexts Are Equal: Matching Interventions to Gameplay

While personalized interventions were highly effective in reducing emotional distress in Simulation Strategy games, their impact

was less pronounced in fast-paced, team-based games like MOBA (Multiplayer Online Battle Arena), where emotional fluctuations were less noticeable. This suggests that the effectiveness of emotional interventions may vary significantly depending on the pacing and structure of the game. In fast-paced team games, emotional changes might be subtler or harder to mitigate through interventions. As seen in figure 9, there was little change in playtime after personalized interventions in such games. For these types of games, exploring alternative intervention modalities, such as subtle tactile or audio cues instead of intrusive text, may be more effective. Further testing across a broader range of game genres is necessary to refine the intervention system for greater applicability.

8.3 Short-Term Emotional Effects and Long-Term Observations

While RageSense reliably reduces negative affect within a gameplay session, our three-week field deployment did not reveal clear evidence of cumulative longer-term reductions in rage levels. To assess whether rage incidents decreased as participants continued using the system, we examined the intervention trigger frequency as a proxy for rage events and computed the Pearson correlation between chronological session order and the number of interventions. The correlation was negligible (Pearson $r = -0.13$), suggesting that intervention frequency did not exhibit any meaningful linear decline throughout the study period. These findings indicate that RageSense provides acute, moment-to-moment emotional stabilization, but such short-term benefits did not aggregate into measurable long-term behavioral change within three weeks. These observations highlight an important area for future investigation: whether longer-term use could cultivate more durable emotional regulation skills beyond immediate gameplay. Longer-term, context-controlled studies will be needed to evaluate these broader effects.

8.4 Privacy Risks and Implications

Although RageSense avoids visual capture, the handling of affective and privacy information warrants careful consideration. Below, we reflect on the privacy risk and implications of our design choices.

8.4.1 Minimization of Affective Data Exposure. Although valence -arousal traces are low-dimensional, they represent sensitive affective information that may implicitly reflect a user's stress level, frustration, or emotional stability. To reduce the exposure of such data, RageSense performs all valence–arousal inference locally on the smartphone, thereby preventing raw ultrasonic reflections or intermediate spectrogram features from leaving the device. Only when a notable emotional fluctuation is detected, a short 5-second valence-arousal history and five obfuscated gameplay screenshots (which are anonymized through the app's built-in Optical Character Recognition (OCR) function that locates text regions and pixelates potentially sensitive strings—e.g., player usernames/IDs, teammate names, and chat messages—thereby reducing identifiability and cross-session linkage) are transmitted for intervention generation. While this design substantially limits the amount of affective data shared, we acknowledge that even minimal emotional traces can carry inference risks, and further reductions remain an important direction for future work.

8.4.2 Identifiability and Characteristics of Ultrasonic Reflections. Prior work [46] has shown that rich skeletal trajectories or fine-grained motion patterns can serve as signatures for user re- identification. Ragesense does capture fine-grained variations in facial micro-movements and device interaction behaviors. However, these reflections encode only non-semantic perturbations in facial geometry and do not contain reconstructible facial appearance or body skeletal trajectories. Nevertheless, we acknowledge that any physiological or behavioral signal may carry residual identifiability risk, and a fuller investigation remains a direction for future work.

8.4.3 Cloud-Assisted Interventions and Practical Trade-Offs. Our current design relies on a cloud-based LLM to generate personalized interventions. While this requires network connectivity, the mechanism is strictly non-blocking: if the connectivity or cloud server is unavailable, gameplay proceeds uninterrupted. As lightweight multimodal on-device models [30, 73, 74] continue to mature, fully local intervention generation may become feasible, further reducing data transmission latency and tightening the system's privacy boundary.

9 Conclusion

This paper presents RageSense, a system for detecting and regulating negative emotions in mobile gaming using near-ultrasonic acoustic sensing. Unlike traditional methods, RageSense avoids visual and audio-based surveillance, offering a non-intrusive, accurate solution for real-time emotion estimation. Paired with personalized, context-aware interventions powered by a large language model (LLM), it effectively reduces negative emotion such as rage, frustration and enhances the gaming experience, particularly in Simulation Strategy games.

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A Application User Interface

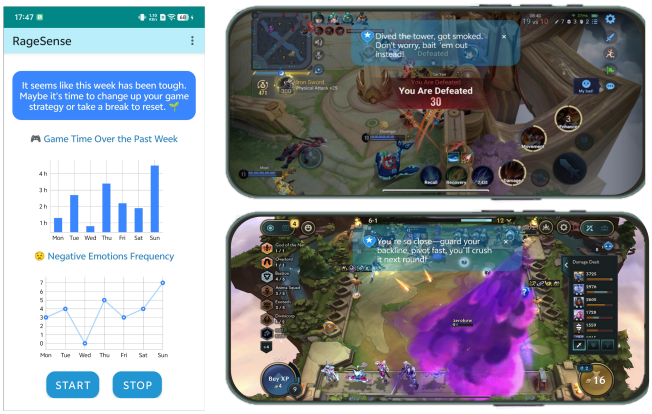

Figure 13: App UI Design. Left: Application UI. Right: Intervention pop-up in mobile gaming.

B Spectrogram Mapping Examples

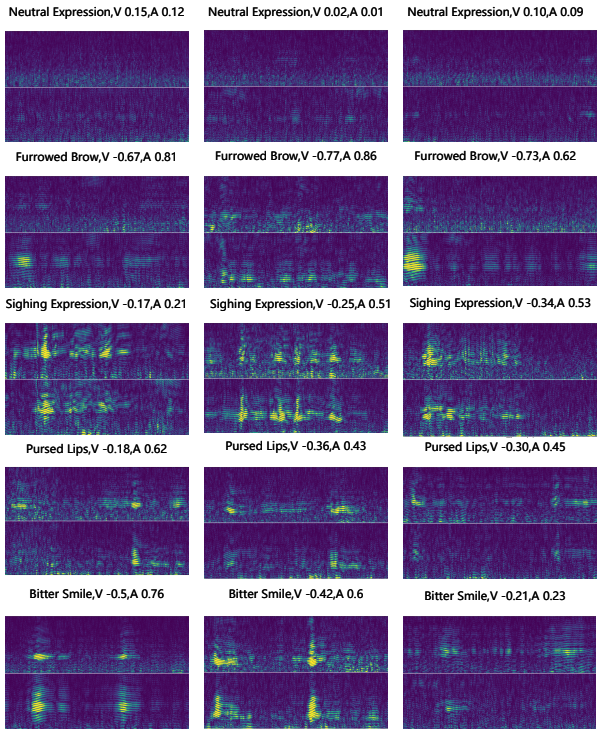

Figure 14: Examples of mapping spectrograms to the Valence - Arousal space.

C JITA Algorithm Pseudocode

Algorithm 1 JITA: EMA + CUSUM + Adaptive Thresholds

Require: Window size N, valence-arousal estimator $\mathcal{M}$, EMA factor α,change-point detector C, dynamic-threshold strategy $\mathcal{D}$
Ensure: Real-time in-app intervention pop-up

Initialize circular buffer $B \leftarrow \text{CircularBuffer}(N)$
Initialize smoothed VA: $V_{\text{ema}} \leftarrow 0,\ A_{\text{ema}} \leftarrow 0$
while gaming session is active **do**
  $S \leftarrow \text{AcquireUltrasoundSpectrogram}()$
  $(V, A) \leftarrow \mathcal{M}(S)$
  $B.\text{append}((V, A))$
  $V_{\text{ema}} \leftarrow \alpha\, V + (1-\alpha)\, V_{\text{ema}}$
  $A_{\text{ema}} \leftarrow \alpha\, A + (1-\alpha)\, A_{\text{ema}}$
  if $|B| < N$ **then**
    continue {wait until buffer is full}
  end if
  $(V_{\text{th}}, A_{\text{th}}) \leftarrow \mathcal{D}(B)$ {Compute dynamic thresholds}
  $cp \leftarrow C(V_{\text{ema}}, A_{\text{ema}})$ {Online CUSUM change-point detection}
  if cp **and** $(V_{\text{ema}} < V_{\text{th}} \ \wedge\ A_{\text{ema}} > A_{\text{th}})$ **then**
    $imgs \leftarrow \text{CaptureScreenshots}()$
    $prompt \leftarrow \text{BuildLLMPrompt}(B, imgs)$
    $intervention \leftarrow \text{QueryCloudLLM}(prompt)$
    if intervention failed **then**
      $intervention \leftarrow \text{DefaultInterventionMessage}()$
    end if
    $\text{DisplayPopup}(intervention, 4\,\text{s})$
  end if
end while